

CH 1 OLTP VS OLAP

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OLTP (Online transaction processing)

OLTP systems are optimized for executing large numbers of small, fast, concurrent transaction while maintaining data integrity through acid properties

every second user performs operation such as

```
login → search product → add to cart → place order
```

each operation involves a few rows only and must complete quickly

Characteristics →

- millions of transaction
- short running queries
- intensive read/write
- high concurrency
- low latency
- ACID compliant
- normalized schema

Eg ATM withdraw, UPI payment, rapido booking

OLAP (Online analytical processing)

OLAP systems are optimized for complex analytical queries over large volumes of historical data

instead of processing individual transaction, patterns are analyzed

Eg

```
SELECT
    State,
    Category,
    SUM(SALES)
FROM Orders
GROUP BY STATE, Category ;
```

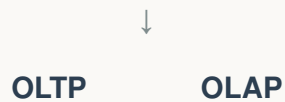
→ *may scan millions of rows*

Characteristics :

- long running queries
- read heavy
- historical data
- for reporting
- star / snowflake schema is common

Eg revenue dashboard, customer segmentation, recommendation engines, quarterly reports

hence companies use multiple databases each optimized for a specific workload (transactional, analytical, caching, search etc)

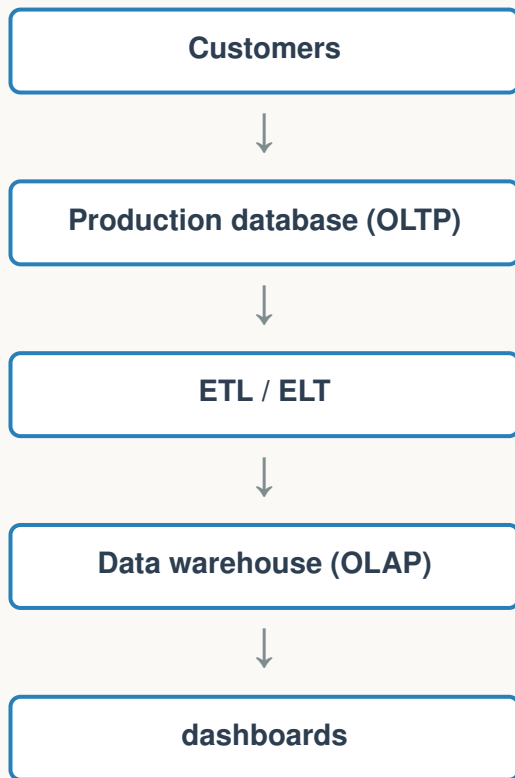


We can't use one database :

let 10000 purchases per minute, now someone runs a complex query over historical data , it causes

- CPU usage spikes
- memory usage increases
- customer transaction slow down , cant place order smoothly

Solution (separate system)



Data warehouse

a centralized repository that stores integrated, analyzed, historical data optimized for report and decision making

OLTP to warehouse

→ done via ETL (extract, transform, load)

- **Extract** → copies the data from relevant fields (Eg Orders, customers, inventory) etc
- **Transform** → clean and prepare the data
 - remove duplicate
 - standardize dates
 - currency conversion
 - eliminate inconsistent data
 - Join related datasets
- **load** → store transformed data in warehouse

OLTP uses normalized schemas to minimize redundancy and ensure consistency during updates hence related fields are linked by foreign keys

OLAP uses denormalized schemas like star or snowflake to reduce joins and improve query performance

OLTP → PostgreSQL, MySQL

OLAP → snowflake, google BigQuery

Practice Questions

Q1. What is the primary goal of an OLTP system? Give three real-world examples.

A1. An OLTP system processes high volumes of short, concurrent transactions while maintaining ACID properties. Examples include placing an Amazon order, withdrawing cash from an ATM, and completing a UPI payment.

Q2. What is OLAP? How does it differ from OLTP?

A2. OLAP systems are designed for analyzing large amounts of historical data using complex queries and aggregations. In contrast, OLTP systems process day-to-day transactional operations with low latency and high concurrency.

Q3. Why is it a bad idea to run large analytical queries on a production OLTP database?

A3. Analytical queries often scan millions or billions of rows, consuming CPU, memory, and I/O resources. Running them on a production OLTP database can slow down or disrupt customer-facing transactions.

Q4. What is a data warehouse, and why do organizations use one?

A4. A data warehouse is a centralized repository optimized for storing historical, integrated data used for reporting, analytics, and business intelligence. It separates analytical workloads from transactional systems to improve overall performance.

Q5. Explain the ETL process.

A5. ETL stands for Extract, Transform, Load. Data is extracted from operational systems, transformed by cleaning and standardizing it, and then loaded into a data warehouse for analytical use.

Q6. A CFO asks for a report showing monthly revenue by product category over the last five years. Should this query run on the OLTP database or the data warehouse? Explain your reasoning.

A6. The query should run on the data warehouse (OLAP system) because it involves historical data, aggregations, and reporting across a long time period. Running it on the OLTP database could negatively affect transactional performance.